Drawbacks of RNN

- RNNs involve sequential computation
  - can’t parallelize = time-consuming
- RNNs “forget” past information
- No explicit modeling of long and short range dependencies

Transformer

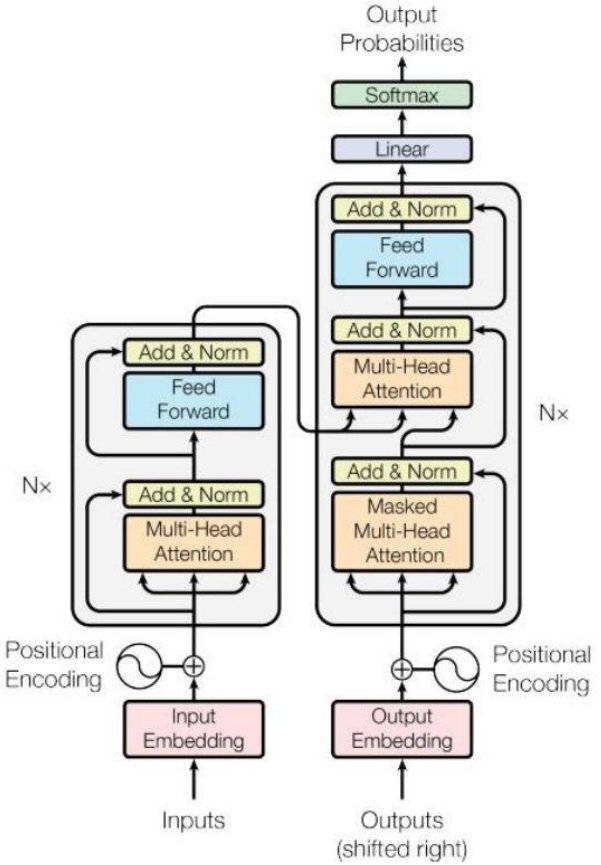
Some slides borrowed from Sarah Wiegreffe at Georgia Tech.

Transformer

“Attention is All You Need”

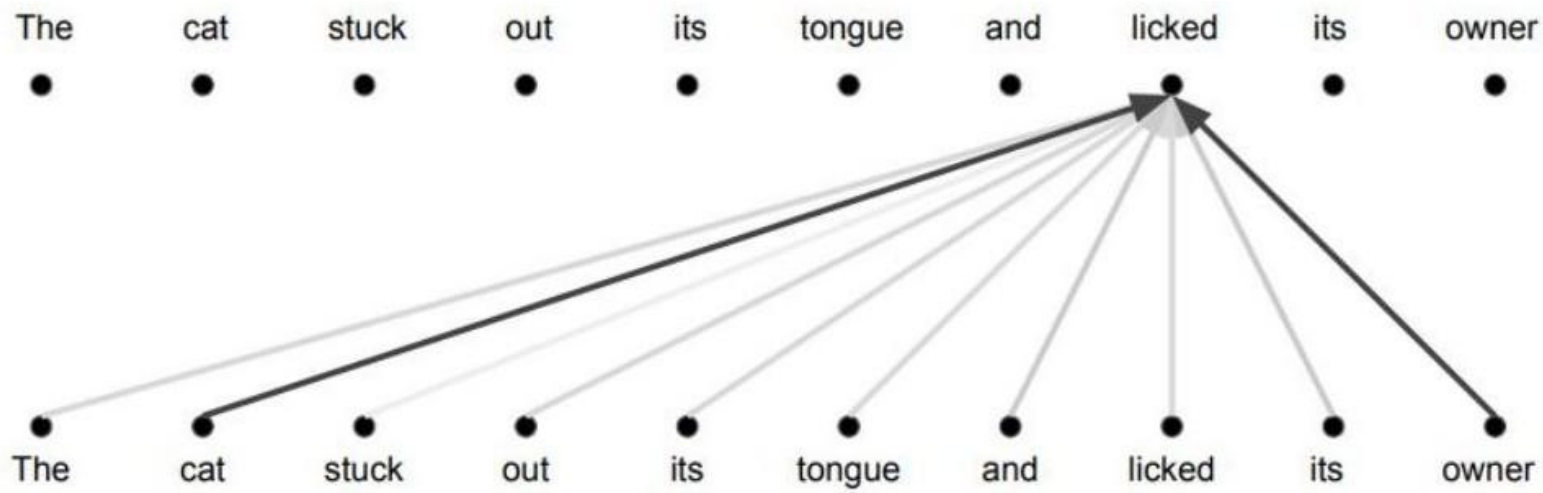
(Vaswani et. al 2017)

Encoder →

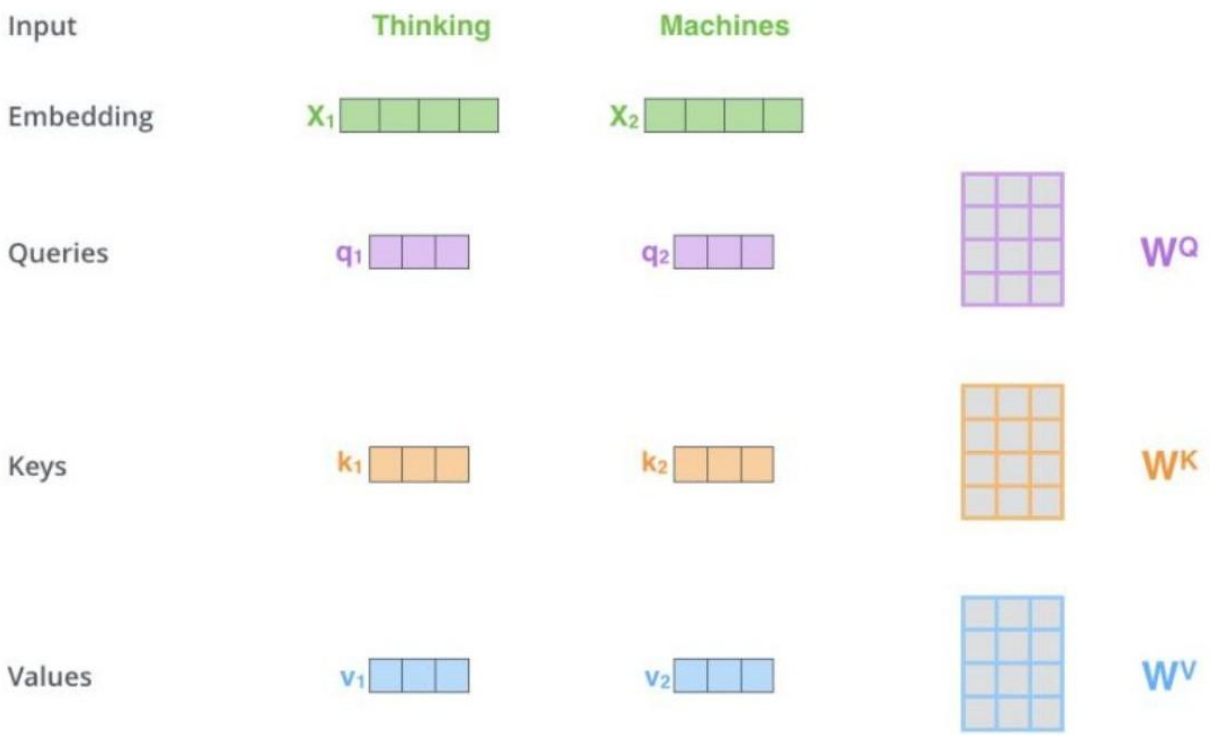


← Decoder

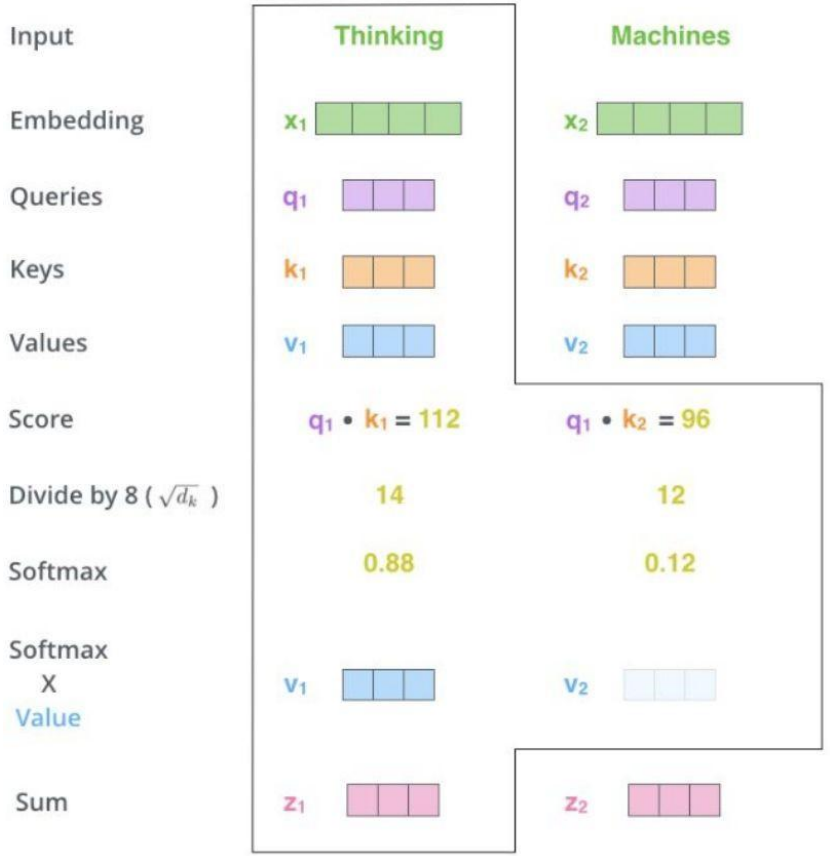
Self-Attention



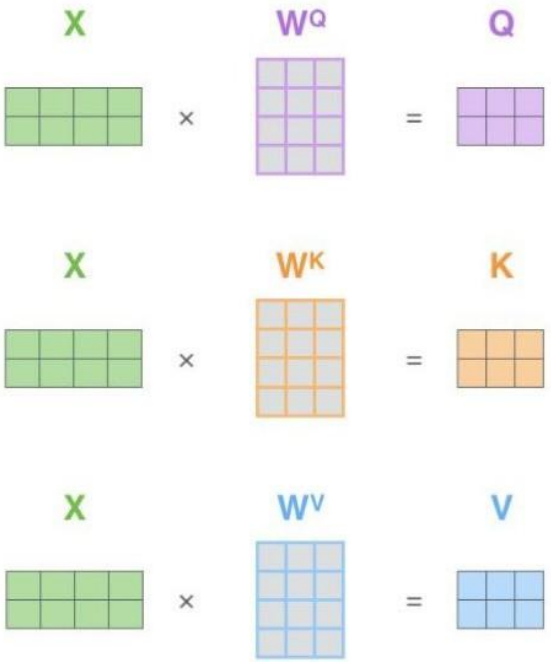
Self-Attention



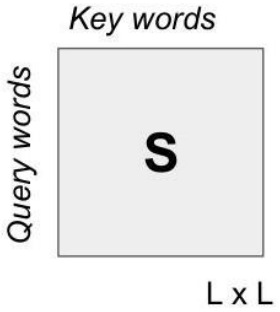
Self-Attention



Self-Attention



$$A(Q,K,V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$



Multi-Head Self-Attention

Parallel attention layers with different linear transformations on input and output.

